UNIVERSITAT D'ANDORRA

Study 12204003 - Bachelor of computer science

Module 8 - Data analysis and project management

Seminar: Project management

Practical exercise Pw2

Ferrocarrils Andorrans

"CargoCare: Two locomotives and one branch line

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1. Introduction

1.1 Antecedents

The calendar shows April 2026. Ferrocarrils Andorrans has completed a successful year 2025. The passenger transport branch has increased its turnover by 30% due to the new tramway offer, named "***Travel as you please***", integrated in our national railway infrastructure.

This year 2026, Ferrocarrils Andorrans has defined a main goal in its strategic plan which concerns the cargo transport activity. The reason for deploying this activity is the state of current cargo services we offer today. Most of the cargo activities are "ad hoc services" realized for a few particular clients in the country. **We want to provide easy to use cargo services for more customers** who deploy distribution of goods within the country. Think about distribution centers and large supermarkets.

The idea to **offer an rail-alternative for transport by trucks** has become a real initiative and has a name: ***CargoCare***. During February and March this year an analysis has been carried out to study the feasibility of this new service. The results, presented beginning this month of April, are promising. As a consequence, we **decided to start a project to realize CargoCare**. Our wish is to **start the service with at least 5 customers from end 2027**.

We must be aware of the magnitude of this macro-project, entitled "**Build CargoCare**". Our vision indicates that we have to define a number of smaller projects which should be synchronized with the main goal: **CargoCare service started**.

This document, describes two projects inside the CargoCare initiative. One is about the purchase of two locomotives for the delivery of wagons to customers and the other is the realization of a branch line at our first CargoCare customer "Juberri Ale".

Fig 1.1: The railway service "Travel as you please" in action.
Credits: https://elcowire.com/urban-rail/

1.2 Next steps

With the decision to realize CargoCare and to have it operation at end 2027, a number of projects have to be defined and planned. The macro-project "Build CargoCare" has already been registered in FA's corporate program, managed continuously with starting beginning June 2026 and forecast delivery at end of year 2027.

In the next chapters we will define the **basics of two projects in the "Build CargoCare" development**:

- Project "**Purchase two shunting locomotives**" necessary to respond to the services, requested by our customers.
- Project "**Build branch line for Juberri Ale**" which makes it possible to serve our first CargoCare customer from the start or the service.

We have to analyze and plan these projects in all its aspects, so we are able to "kick them off" well documented.

1.3 Your mission in a nutshell

The main macro-project will be realized by carrying out individual projects, to be prepared and planned in detail. The involved employees have to be assigned to these projects, applying a reasonable workload policy.

Your mission covers two projects, to be prepared in all its details:

- `FA-COR-FRE-02-CACARE_P2026011`: **Purchase two shunting locomotives**
- `FA-COR-ISC-02-CACARE_P2026012`: **Build branch line for Juberri Ale**

Both projects, one owned by Freight transport group and the other by Infrastructure construction will be explained more in detail in this document.

You, as a team of specialists, will be assigned as project owner of the two projects to develop in this mission.

So up to ***you, project managers***, to prepare the projects in detail. You can count on a group of people, composing the project team to do the job. At the same time, fellow-project managers will execute other projects in our macro-project “***Build CargoCare***”. They need your information about your project planning and progress in order to synchronize their activities with you.

You should plan your two projects in this way that both will be realized as soon as possible and are synchronized where necessary. In other words: **the results you deliver allow both projects to be kicked-off immediately**.

1.4 Available documentation

Table 1.1 shows the available documents, related to this project, which could be useful as additional information:

Table 1.1 - Available documents for the project

Reference	*Title*	*Description*	*Version*
`UDACPC_2026_W_0036`	Practical work 1 Pw1	This document contains a first idea about the service to deploy	`0.3.0`
`UDACPC_2026_W_0037`	Product proposal (solution document for practical work Pw1)	Latest version of the CargoCare service proposal, revised after the presentation for the board and directors	`0.1.0`
`UDACPC_2026_W_0038`	Teacher’s notes (personalized for you)	The personal teacher’s notes about your project preparation in the practical work Pw1	`1.0.0`
`N/A`	Practical work delivery	Your practical work Pw1 delivery, commented in document `UDACPC_2026_W_0038`	`N/A`
`UDACPC_2026_P_0011` to `UDACPC_2026_P_0017`	The course’s documentation	Documents, presented at your business management training	
`UDACPC_2022_P_0021`	practical case “Ferrocarrils Andorrans”	The company’s description and general information about its activities and infrastructure	`0.1.0`

These documents contain the basic information for the projects to realize.

1.5 Steering committee and project team

The steering committee is composed by two members of the board, the commercial director and the teacher who coached the project team to prepare the proposal for this project (that’s me :-)).

The project team will be defined during the project preparation, done by the project manager, assigned to this project and a colleague (that are you :-)). A group of human resources, assigned to this project has already been reserved for the project.

The steering committee does not spend working time in your projects. That’s why we don’t mention these people in the human resources list in chapter 2.2.5.

2. The projects, organization and context

2.1 Overview and relation

The two projects to be prepared and executed are specified in table 2.1

Table 2.1 - The two projects inside macro-project “Establish CargoCare”		
Code		***Title***
Long	***Short***	
FA-COR-FRE-02-CACARE_P2026011	P2026011	Purchase two shunting locomotives
FA-COR-ISC-02-CACARE_P2026012	P2026012	Build branch line for Juberri Ale

The relation between these two projects exists in the **requirement of new delivered locomotives to test and approve the correct functioning of the new branch line**. The knowledge, acquired during the execution of these two projects will be used for future, similar projects.

The two projects will be detailed in chapters 3 and 4 of this document.

2.2 Environmental and contextual elements

2.2.1 Railway infrastructure

The new service will use the existing railway network in Andorra. Traffic control at Andorra la Vella will be prepared for receiving the freight trains which contain the wagons to be delivered and picked up at the CargoCare customers. This activity will be operational during end 2027 and is defined in the corresponding business processes.

2.2.2 Budget

It is assumed that the budged for all projects in macro-project “Establish CargoCare” has been approved and made available. No financial tasks will be performed in our projects.

2.2.3 Working time and holidays

Employees of Ferrocarrils Andorrans have 25 holidays to spend throughout the year. The project teams for both projects `P2026011` and `P2026012` will suspend their activities during the following periods (specified dates included):

- **3d - 21st of August 2026** (Summer 2026 holiday)
- **21st of December 2026 - 6th of January 2027** (Christmas and season’s holidays)
- **22nd - 29th of April 2027** (Easter period)
- **7th - 25th of August 2027** (Summer 2027 holiday)

The 100% working time is **five days a week, 8 hours per day**, so a total of 40 hours a week.

Important:
We assume that the external companies, working for us, don’t interrupt their activities during the FA holidays. However, they have to wait for us during these periods.

2.2.4 Purchase procedure

In our company, **purchasing goods, supplies and services** has been organized according to the following organization:

- A purchase starts when the **project manager notifies the purchase department about the acquisition of products and/or services**. This notification is documented by a document, named “**Request for purchase**” which contains a global description of the products/services to acquire and an estimation of the total cost.
- Purchase department **checks the cost with the financial department if a budget is available**. If not, the project manager will be notified about the rejection of the purchase. This stops the procedure. Otherwise, the project manager will receive the “go-message” and the procedure continues.
- **Purchases with an estimated cost of less than 10.000 Euros** will be done directly and follow another process, **FA-SUP-ABU-02-ACREDP** named “*Acquire resources by direct purchase*”. This process exists, so no need to detail it here.
- The purchase continues with the creation of a document, named “**Request for proposal”** or **RFP.** The document has three main parts:
  - **Technical chapter**: Specification of the products and/or services to acquire
  - **Legal chapter**: Specification of the conditions and rules to apply for the purchase
  - **Financial appendix**: The estimated cost of the purchase, detailed in a calculation.
- In the **Request for proposal**, the requested **articles and/or items will be listed** according to the enumerations, with additional details to avoid ambiguous options.
- If the total **estimated purchase cost is more than 200.000 Euros**, the **board has to authorize the purchase**. This will be decided in the next board meeting which takes place **each first working day of the month**. Used documents: Request for proposal. This step is a go/no go condition.
- All purchases **between 10.000 and 200.000 Euros** (both amounts included) Euros can be launched without authorization, but the board has to be notified.
- For **purchases with an estimated cost of more than 100.000 Euros**, the project manager will now select the candidate providers. He/she will do this by means of a “**Request for information**” or RFI at each candidate provider. The answer has to be r**eceived within 10 natural days** from the moment of sending the request for information document.
- If, for any reason, the project manager **cannot select at least three providers** (using the RFI), he/she will extend the selection procedure for more providers. **This will be repeated until at least three providers have been selected**.
- For **purchases with a cost of 100.000 Euro or less, the RFI procedure is not necessary** and the RFP will be sent directly to as many candidate providers as possible.
- The **Request for proposal** (without financial appendix) will be sent to the **candidate providers** which can be national or international. This depends on the availability but national providers have preference.
- The candidate’s **proposals must be received before a clear, defined due date**, which is **maximum 30 natural (calendar) days** from the moment of sending the request to the providers by e-mail. This period can be shorter for less important purchases.
- After the **due date** has been arrived, all **proposals will be opened at the same moment in a formal meeting** (maximum two days after proposal closure) at a predefined place. All candidate providers who have issued an offer are welcome to be present and will be invited.
- Once **opened**, Ferrocarrils Andorrans has a maximum of **15 working days to select the provider**. The main criteria is the price, but elements like the nature, the delivery delay and the quality can be used to select the final provider. The decision must be documented in a “**Purchase Proposal**” which must be a**pproved by the board for orders from 100.000 Euros** (during the next board meeting). Lesser orders can be directly processed.
- After the **selection has been closed**, and, if this applies, the board has granted authorization, the **decision will be communicated** to all candidates.
- The **selected provider** will **receive a purchase order within five working days** after the communication of the decision.

- The **selected provider confirms the purchase** order and **communicates a planned delivery date** of the products or services.
- The chosen **provider now starts the delivery procedure**, to be able to respect the delivery delays of all ordered articles.
- Once delivered, **FA accepts the delivery** in an official **acceptance session**. See this session as a sub-process. Eventual problems will be noted and the decision will be made whether the order is **accepted**, **accepted with issues** or **rejected**. The provider shall correct eventual issues within 10 natural days.

The **actors** in this procedure are:

- **Board**: Authorization of requests for proposal and purchase orders.
- **Financial department**: Purchase cost estimations
- **Purchase department**: Legal affairs, Proposal reception, Purchase order and proposal opening event
- **Project management**: Technical chapter of request for proposal, technical contact with candidate providers and provider selection.
- **Macro-process owners**: Delivery acceptance session (ex: Rolling stock group for the acceptance of delivered locomotives).
- **Providers**: Communication of requests for proposal, proposals, purchase orders and delivery activities. This actor is “loosely linked” to our purchase procedure.

Although this procedure has been documented in words, a **formal business process model does not exis**t. Nevertheless, the formal process code and title has been reserved:

`FA-SUP-ABU-02-PURCHA - Purchase products and services`

See the **Macro Process Map of Ferrocarrils Andorrans** to place this important business process in the company’s context.

As a consequence, the **creation of UML-EP and BPMN-2 models for this business process is one of the tasks to deploy in your mission**.

2.2.5 The project team. Human Resources for your two projects

You can count on a team for the two projects, detailed in table 2.2.

Table 2.2 - Project team resources			
Resource code	***Title***	***Main tasks in the project***	***FTE*** [1]
RHMNG	Project management	The project owners for the two projects	2
RHFIN	Financial support	Cost estimations, control and accounting	1
RHPUR	Purchase department	Purchase of the products, used in the projects	3
RHCON	Construction group	All construction activities, like platform building, electricity, networks, lightning and similar tasks	6
RHTRA	Track group	Track laying, including turnouts, signals and other related infrastructure	6
RHITG	Information technology group	Installation and maintenance of IS/IT network and communication infrastructures, excluding the cables (done by construction group)	1
RHDRV	Rolling stock group	Train drivers, eg. For shunting locomotives.	2
RHTRC	Traffic control	Configuration of test drives after branch line construction.	1
RHTRU	Truck drivers	Transport of all materials over the road, including loading and unloading, using trucks with integrated cranes	2
RHCOM	Commercial group	Relation with customers, instances and press, eg. To communicate activities	1
RHRRH	Human resources	Training activities, employee assignment	1
RHQUA	Quality group	Testing and acceptance of all installations	3
		Total FTE:	**29**

These resources will be available to realize the two projects, explained in the next chapters. The available people are given in **FTE**[1]. Assign the people in this way that the project can be realized in the shortest time. **Individually employees must be assigned with the prefix and a sequence number**. Ex: **RHDRV2** for truck driver 2.

The board has expressed that the date to **launch the CargoCare service should be the 1st of December 2027**. Our two projects should be ready before this date.

About resource RHMNG - “Project management”: This resource will dedicate 100% of its time to deploy the following tasks:

- Kick-off meeting (1 day)
- KPI calculation and update (0,4 days for the duration of the project)
- End report (during last phase, 3 days)

Typically, the people who prepared the projects in this mission will be the project owner. The activities should be spread out over the project’s execution period.

[1] FTE means “**Full Time Equivalent**”. It is the number of people who can work full-time in activities. Example: 2 people, working half of the time represent 1 FTE.

3. Project P2026011 - Purchase locomotives

3.1 The project’s title and its goal

The title for the project to realize has be chosen as follows:

Purchase two shunting locomotives

And its goal is:

Acquiring two shunting locomotives for the existing Andorran railway network, adapted to the future service “CargoCare”, following the formal corporate purchase business process, including transport by railway.

3.2 Calendar and kick-off date

This project is one of the first activities of macro-project “CargoCare”.

In the CargoCare project group, we have decided to **start this project Monday, the 1st of June 2026**. A kick-off meeting must be held at this date.

3.3 What must be realized?

The project aims for the acquisition of additional shunting locomotives in a relatively short period. These locos will be used to test the branch line at Juberri Ale to be build in project `FA-COR-ISC-02-CACARE_P2026012` - “Build branch line for Juberri Ale”. Hereafter, these locos will be used in the delivery of cargo process of service “CargoCare”.

Figure 3.1 - A typical shunting diesel-loc at work. Notice the absence of a catenary at the branch line.

3.4 Purchase procedure

The **purchase planning follows exactly the organization, explained in chapter 2.2.4 - Purchase procedure**. See this project as a single instance of process
`FA-SUP-ABU-02-PURCHA - Purchase products and services`,
adapted to the specific circumstances for this purchase.

3.5 Preliminary information

The following facts have already been obtained during internal investigation:

- The estimated **order amount** for the two locomotives is about **810.000 Euros**
- The minimum **delivery delay** for a locomotive car is **55 natural days**. No holiday period will be taken into account.
- The **transport** of the locomotives takes **5 days** and will be done by rail to arrive at the loc shed at Andorra la Vella

This info must be taken into account in the project planning.

The locomotives, we are looking for could be second hand or new products. Anyhow, we will buy the vehicles and not lease them.

See figure 3.1 on the previous page for a typical example of a shunting locomotive.

3.6 The project’s main activities

3.6.1 Overview

Table 3.1 shows an overview of the activities, to be deploys in this project. The codes have a length of 6 characters, including the prefix “**PSL**” (**P**urchase **S**hunting **L**ocomotives)

Table 3.1 - Activities for project “Purchase two shunting locomotives”

Activity code	*Title*	*Description*	*Estimated effort (days)*
PSLPRE	Preparation	Creating the final version of the planning and synchronization into the CargoCare group.	**4**
PSLRFI	Request for information	The request for information about shunting locomotives at at least 6 providers in Europe.	**3**
PSLRFP	Request for proposal	All documents, necessary to launch the RFP procedure, including the Board’s authorization, the candidate provider selection and sending the requests to the providers.	**9**
PSLPAN	Proposal analysis	Reception of the proposals and official opening event, followed by the analysis.	**7**
PSLSEL	Provider selection	Selection and creation of purchase proposal. Includes the authorization of the Board.	**3**
PSLORD	Purchase order	Creation and communication of order and result to other candidates.	3
PSLOFU	Order follow-up	Follow-up of the order realization by weekly meetings with the provider	8
PSLDEL	Delivery	Locomotive reception and acceptance, documented in a formal acceptance report	5
PSLTRN	Training	Training for four new locomotive drivers.	5
PSLCLO	Closure	Final KPI calculation and end reporting	4
		Total effort:	51 days

3.6.2 Activity PSLPRE - Preparation

The following activities must developed:

- The creation of the **final version of the plannin**g
- The final **revision of the SWOT analysis**
- The **kick-off meeting**, scheduled at the 1st of June 2026, where the results of the previous two activities are presented in a formal kick-off presentation.

The estimated effort could be divided equally over these activities

3.6.3 Activity PSLRFI - Request for information

The effort consists of:

- Finding **possible providers** for shunting locomotives: 1 day.
- Creating a **formal inquiry to candidate providers** in order to know whether they are able to deliver us the shunting locomotive in the estimated delivery delay: 1 day.
- **Sending the inquiry document** to the candidates: 0,5 day.
- **Analysis of the reactions and final selection** of the candidate list. 0,5 day.

This activity must start 1 day after the realization of the kick-off meeting in activity **PSLPRE**.

3.6.4 Activity PSLRFP - Request for proposal or RFP

Several documents must be created here by different groups. See the human resources list, chapter 2.2.5 , to select the right groups.

We estimate the following effort for each document:

- Technical part: 3 days, typically done by the project manager
- Legal part: 1 day (this is a standard document)
- Financial analysis: 2 days.

The preparation for the Board’s authorization request will take another day. **Attention**: You have to wait for the board’s authorization in order to continue this activity.

Then, to finish this activity, the following tasks must be carried out:

- Candidate provider **selection** (2 days)
- **Sending RFPs** to the providers (purchase dept., 1 day)

Once sent the RFP to the candidates, we allow them a **period of 15 natural days** to respond with a proposal.

3.6.5 Activity PSLPAN - Proposal analysis

Once received, the proposals must be opened in an official meeting where the candidate providers are invited. Each offer will be opened and the total price will be communicated.

Once opened, the proposals will be analyzed in order to obtain all elements for taking the decision in the next activity.

The organization of the opening event, done by the purchase department, takes two days. The analysis will take about 5 days more.

For this RFP, the **analysis and selection period will be 7 working days**.

3.6.6 Activity PSLSEL - Provider selection

Once analyzed, the final provider will be selected. The decision will be documented in a formal report, being the purchase proposal. Because the total order amount will largely exceed 100.000 Euros, this proposal must be authorized by the Board. Only then, an official purchase order can be processed and send to the selected provider.

This activity takes about two days of effort. The authorization will take place the next Board meeting. We assume that the purchase will be authorized without any delay.

Attention: Because of the holiday period from 3 - 21 August 2026, the Board will **advance the August’s meeting to Thursday, the 30th of July 2026**.

3.6.7 Activity PSLORD - Purchase order

Once authorized, purchase department will deploy two tasks:

- Sending the purchase order to the selected provider (2 days effort)
- Informing the other candidates about the disapproval, thanking them for the service. (1 day)

3.6.8 Activity PSLOFU - Order follow-up

Once sent the order, the **project manager will hold periodic meetings** which serve to guide the order realization. These meetings will take place mostly on-site at the provider and take **two days each week** during the delivery delay.

If this period corresponds with the summer holidays, the meeting will be held before and after this period.

3.6.9 Activity PSLDEL - Delivery

Once created, the locomotives will be **delivered both at the same time**. Take into account that after the delay, 5 days must be included for transport, done by the provider.

Once arrived, the locomotives have to be accepted by Ferrocarrils Andorrans one by one..The rolling stock group must be involved in this task. Each locomotive needs two days of effort to be accepted and another four days should be planned to resolve and verify possible issues to be fixed. The project manager has to be present at least one day per locomotive to authorize its acceptance.

3.6.10 Activity PSLTRN - Training

Five engineers will be trained for driving the new shunting locomotives. The organization of this event is done by the human resources group and takes 5 days effort.

You can plan this activity only for human resources. The actual training session is outside our project and is an established business process.

3.6.11 Activity PSLCLO - Closure

This activity will be deployed at end of the project and contains the following tasks:

- **KPI final calculation** and interpretation (1 day)
- **End-report** creation (2 days)
- **Debriefing meeting** with CargoCare macro-project group (1 day)

Once done, the project will be closed and the locomotives will be officially transferred to the Rolling Stock group.

4. Project P2026012 - Build branch line

4.1 The project’s title and its goal

The title for the project to realize has be chosen as follows:

Build branch line for Juberri Ale

And its goal is:

Construction of a branch line for Customer “Juberri Ale” at Sant Julià de Lòria, connected to the main line Andorra - Sant Julià which makes it possible to deliver and pick up wagons between our shunting yard and the brewery Juberri Ale.

4.2 Calendar and kick-off date

This project will create the infrastructure for our first CargoCare customer. In the TCargoCare project group, we have decided to **start this project Tuesday, the 9th of June 2026**. A kick-off meeting must be held at this date.

4.3 Current situation

Figure 4.1 shows the current situation. The Juberri Ale brewery has a connection to the public road and (un)loading activities take place at a terrain before the brewery building. Not far from this, the double mainline track Andorra la Vella - Sant Julià de Lòria is positioned: Track east has code **`AE`** and track west is identified by code **`AW`**. This railway has a catenary having the standard 25 kV, 50 Hz AC voltage.

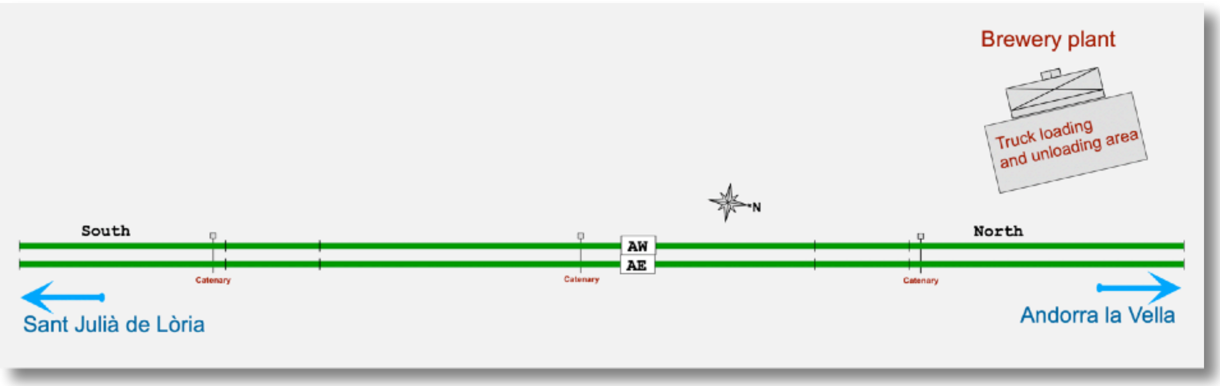

Figure. 4.1 - Current situation of tracks and infrastructure in the Juberri Ale neighborhood

The original graphic file for this figure can be found in the document map in folder “`trackplans/elements`” as file `CargoCare_JuberriAleBranchLine_Current`.

The situation is ideal for the construction of a branch line between main line and the brewery. Delivery and pick-up of wagons will be done from the shunting yard in Andorra la Vella, so consists will always arrive from the north.

An important factor is the train traffic regulation: **In normal operation, trains will always use the right track of a double track railway**. The **left track is only used in exceptional situations**, eg. During maintenance or construction works.

4.4 What must be realized?

The project includes the **construction of a full-functional branch line** between the brewery’s loading facilities and the main line Andorra - Sant Julià, including two secured crossover turnouts in the main line and an arrival track to **allow delivery and pick-up of wagons in one, single shunting process instance**.

Figure 4.2 shows the final situation with branch line connection, tracks to the brewery and loading and unloading infrastructure. **For clarity, all elements are coded**. We ask the project team to use these codes in the project model and planning. The original drawing is available in the “`trackplans/elements`” folder as file `CargoCare_JuberriAleBranchLine_New`.

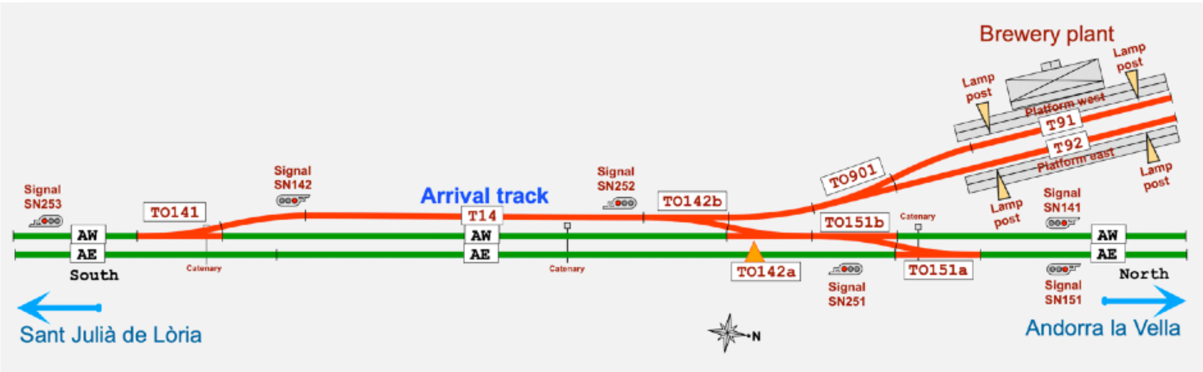

Figure 4.2 - Projected branch line and infrastructures for Juberri Ale

The new **tracks to be realized as an extension are marked in red**. Notice also the track and turnout codes. The main line is coded with two capitals (ex: `AW` and `AE`), the sidings have numbers with prefix `T` and the turnouts are coded `TO` and a 3-digit number.

Ferrocarrils Andorrans will lay the tracks, the turnouts and construct the infrastructures around the branch line. This includes the following elements:

- The **track work,** including all turnouts and tracks.
- All **signals**, including the connection and integration in the Ferrocarrils Andorrans signaling system.
- Two **platforms**, made of concrete, situated at the current truck loading and unloading facility.
- Four **lamp posts**, two for each platform, to be placed at the same time as the platform construction

Paragraph 4.7 - “Project’s main activities” will enumerate the activities in detail.

So, the following elements will not be excluded from this project,:

- **New catenary**: The shunting locomotives will use a “**last mile” option**, being a diesel motor which gives the loc a hybrid nature.

- **Access roads:** We assume to have access to the construction site by using existent roads and paths. No need to plan work for this item.

4.5 Some basic information about tracks, signals and infrastructure

Turnouts:
These track elements with code “`TOnnn`” allow a train or tram to change from one track to another. The position can be changed from left to right with a motor, commanded at traffic control. Manual turnouts are mainly used at shunting sites. For the Juberri Ale branch line, we have to install a total of 6 turnouts.

Detail: In British English a turnout has the name “Point”.

Catenary:
This is a overhead wire which powers electric locomotives and other track vehicles which are equipped with pantographs”. In our project, the catenary will not be extended. See figure 4.3 for details.

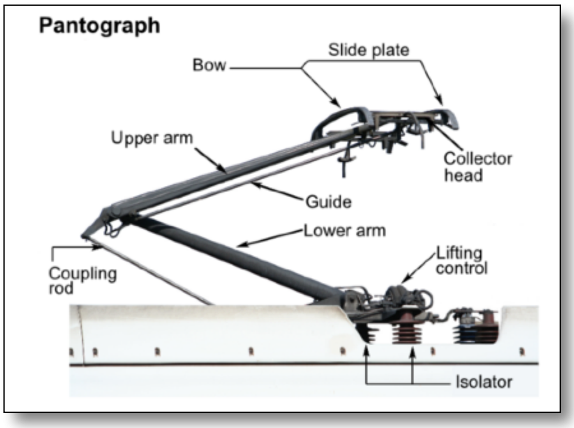

Figure 4.3 - Anatomy of a pantograph.
Source: Wikipedia

Platforms:
Concrete base at a level which allows load and unload of wagons with forklifts. The two platforms at the brewery have a length of 50 meter, which allows 4 wagons of type “**HBis**” to be placed at each platform. See figure 4.4 for a photo of such a wagon.

Signal:
The six train signals SN141, 142, 151, 251, 252 and 253 have to control the arrival and departure of a consist with wagons for the brewery. Other signals, installed farther away, will control the transit of other trains and are not included in our project.
A signal has an aspect, which can be one of the following:

- **Red**: **Stop!**
- **Yellow**: Proceed, but prepare to stop at next signal
- **Green**: Proceed.

Figure 4.4 - An example of a 2 axle wagon of type “HBis” from “RENFE”

All signals will be controlled by FA’s traffic control.

Thermite:
Thermite is used to weld tracks in-site. It is a chemical mixture of Iron oxide (Fe_2O_3) and Aluminum (Al) which is ignited to start the following reaction:

$$Fe_2O_3 + 2Al \rightarrow 2Fe + Al_2O_3 \text{ + heat and light}$$

This highly exothermic reaction produces liquid iron, caused by the redox-reaction of the iron oxide with the metallic aluminum. This liquid iron sets in the gap between the two tracks to weld.

For a demonstration: Se this video from Wolfgang Lendner.

Turnout heating system:
This is an equipment to de-frost the turnouts and to remove accumulated snow from the moving parts. It is simply done by heating the turnout “frog”, motor and “blades” with a gas-flame. The system can be remotely controlled from traffic control. This infrastructure is excluded from our project and will be installed later.

4.6 Shunting process

The projected branch line and arrival track have been defined after a shunting process analysis, which will be explained in this chapter.

Looking at figure 4.2, you notice that all turnouts, tracks and signals are formally coded. We will use these codes in the shunting process, which takes into account the following conditions:

- **Delivery and pick up** of wagons must be possible in **one, single process instance**.
- All **hauling and placement of wagons** will be done by FA’s shunting locomotive. Juberri Ale does not have any traction power on site.
- **Juberri Ale loads and unloads the wagons.** This can be done equally at both platforms east and west.
- **Ferrocarrils Andorrans controls the traffic** with the installed signals **until arrival at arrival track T14**.
- All **traffic between T14 and both platform tracks T91 and T92** will be **coordinated by personal of Juberri Ale**.
- **Departure of the consist** with picked up wagons to the shunting yard is controlled by **Ferrocarrils Andorrans traffic control.**
- A **shunting process will only be performed** after the **CargoCare delivery schedule has been signed** by Juberri Ale.

Said this, a delivery/pick up instance will be performed according following these steps:

1. The shunting locomotive arrives with maximum of four wagons (or “consist”) at the north bound track **AW** at signal **SN141**.
2. When signal **SN141** indicates “Proceed”, the consist passes turnouts **TO151b**, **TO142a** and **TO142b** to arrive at arrival track **T14**.
3. The engineer of the consist announces its arrival at the personal of Juberri Ale, who grants access to the branch line, setting the turnout **TO901** to give access to the empty track. This can be **T91** or **T92**.
4. The consist will enter the branch line by pushing the wagons to be delivered to the empty track.
5. The delivered wagons will be uncoupled and the locomotive will move beyond turnout **TO901**. Which will be changed to give access to the track with the wagons to pick up.
6. The locomotive will go to these wagons and couple them.
7. The new consist with the picked up wagons moves to the arrival track **T14**.
8. The locomotive will be uncoupled and waits for the admission to go to track **AW**, south bound, past turnout **TO141**, and signal **SN253**, both controlled by FA’s traffic control. The locomotive changes direction.
9. When signal **TO141** shows “proceed” the locomotive goes to track **AW**, northbound, past signal **SN141** and changes direction.
10. When signal **SN141** shows “proceed” the locomotive goes to the wagons at the arrival track and couples them. The brake-pipe will be pressurized and a brake test will be performed.
11. The engineer of the new consist communicates the “ready for departure” to FA traffic control and waits for further instructions.
12. When signal **SN252** shows “proceed”, the consist with picked up wagons returns to the shunting yard via turnouts **TO142b**, **TO142a**, **TO151b** and **TO151a** and track **AE**, northbound.

Of course, we could model this using BPMN-2, but we have limited time and have to prepare the CargoCare projects.

4.7 Available materials

The following **materials are already in stock at Ferrocarrils Andorrans shunting yard** and don’t require a purchase:

- **Track and turnouts**, included all materials to construct and install these elements.
- **Train signals** to secure the branch line.
- Materials to build the **platforms**, including the **lamp posts**
- All **materials** to **install electricity** and computer networking

Nevertheless, you should plan the transport of the materials from the shunting yard to the site, by railway, when needed.

The concrete, required to build the platforms has to be ordered at a local factory. More information about this in the detailed project description.

4.8 Project main activities

4.8.1 Overview

Table 4.1 shows an overview of the activities, to be deploys in this project. The codes have a length of 6 characters, including the prefix “**BJA**” (**B**ranch Line **J**uberri **A**le)

Table 4.1 - Activities for project “Build branch line for Juberri Ale”

Activity code	*Title*	*Description*	*Estimated effort (days)*
`BJAPRE`	Preparation	Revision of the final version of the planning, including definition of KPI2 and and synchronization into the CargoCare macro-project group	**4**
`BJASIP`	Site preparation	Preparation of the terrain to build the tracks, realized by the construction group	**11**
`BJATRB`	Track building	Integration of the turnouts and laying track for the arrival track `T14`. Installation of all signals	**23**
`BJAPLB`	Platform building	Realization of the two platforms, using concrete, including the installation of the electricity and lamp posts.	**22**
`BJATST`	Test drive	Integral test of the entire installation using the two new shunting locomotives, acquired in project `FA-COR-FRE-02-CACARE_P2026011`	**6**
`BJACOM`	Communications	Public relations activity to communicate and broadcast the branch line realization in collaboration with RTVA.	**4**
`BJACLO`	Closure	Final KPI calculation and end reporting	**6**
		Total effort:	`76 days`

The mentioned main tasks will be developed more in the next sub-chapters 4.8.2 - 4.8.8.

Notice that project management has been described in chapter 2.2.5 - Human Resources. This activity must be included in the project planning.

4.8.2 Activity BJAPRE - Preparation

The following activities must developed:

- The creation of the final version of the planning
- Definition of all KPIs for the project.
- The final revision of the SWOT analysis
- The kick-off meeting, scheduled at the 9th of June 2026, where the results of the previous two activities are presented in a formal kick-off presentation.

The scheduled 4 day effort could be divided equally over these activities.

4.8.3 Activity BJASIP - Site preparation

Before starting any track construction activity, the site must be prepared and secured. This includes the following tasks (in order of execution):

- Securing the railway traffic at the site for passing trains with formal warning signs. See figure 3.2. (2 days). The signs are available at Ferrocarrils Andorrans.
- Transport gravel and sand to the site (2 days), done by the transport people.
- Removal of vegetation. (3 days)
- Land leveling and conditioning, using gravel and sand (2 days)
- The construction of a temporary gravel road which gives access to the site (3 days)

The estimated effort is specified for each task. This activity has priority and must be finished before any subsequent construction of the tracks.

Fig. 4.4 - Security system for passing trains during track construction
source: https://www.spitzke.com/en/home/

The following constraints must be taken into account:

- This activity **cannot start before the 1st of July 2026**.
- It must be **finished within 8 working days**.
- **No work can be developed without the installed security**. Once installed, **allow 24 hours before starting** other tasks, so the regular train drivers will adapt themselves to the new situation
- The **transport of all materials** will be done by the truck drivers. We assume that trucks will be available without limitation.

4.8.4 Activity BJATRB - Track building

The tracks to be build require to interrupt the regular traffic temporarily. Thus, this activity requires more detail, explained in this chapter.

The realization of track modifications must respect the following rules:

- **The traffic must never be interrupted totally**. Ferrocarrils Andorrans allows the installation of **alternate traffic for a maximum of two days** (or 48 hours).
- At the **end of period of alternate traffic**, **normal traffic must be possible for a period of at least 3 working days** before installing a new interruption.
- **Before start of a track building** activity, the **weather forecast must be consulted** at the official Andorran meteorological institute, who issues an advice. **In case of a negative advice** due to severe weather, the **track building activity must be postponed a maximum of 3 working days**.

As a consequence, the **turnouts in the main line must be installed in a certain order**, organized in two phases: West and East.

Taking this into account, we have to plan the track building activity the following way:

1. Install alternate traffic, using west bound track **`AW`**.
2. Install turnout **`TO151a`** in the main track **`AE`**. Lock position to “straight”
3. Resume normal traffic for 3 working days.
4. Install alternate traffic, using east bound track **`AE`**.
5. Install turnouts **`TO151b`**, **`TO142a`** and **`TO141`** in main track **`AW`**. Lock positions of turnouts **`TO142a`** and **`TO141`**. Take **`TO151a+b`** into service.
6. Resume normal traffic.
7. Build the rest of the branch line. This does not affect the regular train traffic.

Following these instructions, we enumerate the **tasks**, with **estimated effort**, for this activity:

- Integration of **one turnout** in the main line: 1 day.
- Laying the arrival track, including turnout TO142b: 4 days.
- Construction of the loading tracks T91 and T92, together with manual turnout TO901: 5 days.
- Placing and connection one signal: 1 day.
- Testing of all elements: 2 days.

The following **constraints** apply to this activity:

- This activity **cannot be started before the completion of activities BJASIP** - Site preparation and **BJAPLB** - Platform building
- Remember: **Only one single track (AE or AW) can be taken out of service at a time** for integration of the two turnouts. The other has to be available to allow trains to pass in a reduced, single-track traffic.
- The **closure of a track is effective only during turnout integration**. The other siding elements can be constructed during full operation.

4.8.5 Activity BJAPLB - Platform building

The two platforms for both stop sides will be built by pouring concrete into a prepared, wooden formwork, armed with steel mats, creating a reinforced concrete surface. Point of attention is ordering concrete at our usual provider (no purchase order needed).

The platforms will be created with stairs and ramps for easy access, all made with the same concrete. Think about preparing the holes for the lamp posts.

For **each platform**, the following tasks will be realized for this activity:

- **Transport** of all materials, like timber for formwork, steel mats to the site (1 day)
- **Order concrete** at our usual provider (1 day), followed by concrete delivery (milestone)
- **Creation of the formwork** and placing the steel mats (3 days)
- Concrete **pouring** (1 day)
- **Formwork removal** after the concrete is set (2 days)
- Placing and connecting the lamp posts: 0,5 day for each unit.
- **Completion**. Several small tasks to complete, not detailed (2 days)

The following constraints exist for this activity:

- **Transport** of the necessary materials takes place over the road with a truck. The truck has capacity for one, single platform, including the lamp posts.
- You could try to **build the platforms by partially overlapping the activities**, but this is not a must.
- The **concrete** must be ordered 3 working days before delivery at site. Order it separately for the two platforms.
- **A concrete truck only can transport enough concrete for one, single platform**.
- The **minimum time between two concrete deliveries** is two working days. We assume to obtain all concrete from the same provider in order to avoid differences in quality.
- Once poured, the **concrete needs 4 days to solidify** enough to allow the formwork to be removed and the lamp posts to be placed.

4.8.6 Activity BJATST - Test drive

At completion of the branch line, the CargoCare shunting process must be tested and validated. This will be carried out with both new shunting locomotives, purchased in the previous project `FA-FA-COR-FRE-02-CACARE_P2026011`: **Purchase two shunting locomotives**.

Be aware that the shunting locomotives can only be driven by trained drivers.

The total test scenario will contain **4 CargoCare delivery cycles with 6 wagons**, in the following scenario:

- **Test drive 1**: Delivery of four wagons with locomotive 1, no pick up of wagons.
- **Test drive 2**: Delivery of one wagon and pick up of the four wagons of test drive 1 with locomotive 2.
- **Test drive 3**: Delivery of one wagon and pick up one wagon with locomotive 2.
- **Test drive 4**: Delivery of 3 wagons and pick up of one wagon with locomotive 1.

Each test drive must be realized from start to end point, so from Andorra la Vella shunting yard to Juberri Ale branch line.

Each test drive takes one day effort form FA traffic control and a train driver. FA traffic control must prepare this test sequence 10 days before, which takes 1 day of work. Between each test drive, at least one day must pass without test activities. In this way, the test results can be analyzed and eventual inconvenient can be fixed. The working time for this latter activity can be omitted in the project planning.

4.8.7 Activity BJACOM - Communications

The activities, performed in this project deserve to be communicated to the public! We propose the following tasks to realize this by public relations:

- **Press conference** at start of the branch line track laying. One day preparation to perform this event the next day (so 2 days work)
- **Reportage**, realized by RTVA (Andorran television) with registration of the 3d test drive and broadcast at the evening hours. During the tests, TV reporters will be present to record the test activities. (2 days work)

4.8.8 Activity BJACLO - Closure

This activity will be deployed at end of the project and contains the following tasks:

- **KPI final calculation** and interpretation (2 days)
- **End-report** creation (3 days)
- **Debriefing meeting** with CargoCare macro-project group (1 day)

Once done, the project will be closed and the branch line will be officially opened bt Juberri Ale.

5. What Ferrocarrils Andorrans asks you to do

5.1 In general

You, as the project’s owner and future **project managers for the two projects** will be asked to **prepare, analyze and plan both projects**.

In this special case, **FA does not require a complete project model**. You have a solid experience in performing this kind of projects, so basic elements are sufficient in order to prepare this project the right way.

There is one issue which you should solve as an extra task: there is no official **BPMN-2 model for the purchase procedure**., explained in words in chapter 2.4.4. FA should be pleased with your proposal for a detailed process model, to be presented to the executive directors of Ferrocarrils Andorrans.

Finally, **your solution will serve to launch the two projects** as a start of the macro-project “**Build CargoCare”**. As a consequence, **create a result which is complete enough to start the challenge**.

5.2 What must be delivered?

Table 5.1 specifies the elements to deliver and the score you can obtain for each of them.

Table 5.1 - Practical work elements to deliver

Code	***Description***	***Value*** *(points)*
Pw201	A UML Eriksson-Penker diagram for the **purchase procedure, described in chapter 2.2.4** (yEd file, pdf export)	5
Pw202	A **Detailed BPMN-2 diagram for the purchase procedure, described in chapter 2.2.4** (yEd file, pdf export)	15
Pw203	A **SWOT diagram** for both projects **P2026011** - ***Purchase two shunting locomotives*** and **P2026012** - ***Build branch line for Juberri Ale***, based on the description in this document. You should mention at least four items per quadrant (pdf file)	5
Pw204	One, single **detailed GANTT chart** for both projects, using **phases**, **milestones**, **tasks** and **dependencies** and visible **human resources assignments**, without overload. If applied, indicate where you apply **float and slack** to create some flexibility. This can be done with the notes option. (GanttProject file)	30
Pw205	Work out **3 KPIs for each project** you will use to measure the project realization. Give each KPI a code. Indicate if it are final or intermediate KPIs. Specify how to calculate each KPI and assign reference values. (spreadsheet like excel or numbers or pdf file).	10
Pw206	An **eye-catching presentation before the CargoCare project group**. You will have **30 minutes** to present the results produced by your team. The **presentation must contain an introduction and a conclusion**. In the **introduction**, you have to present the **team which worked on this project preparation** and the **role of each member**. In your **conclusion**, Indicate and explain clearly the end date of both projects, referring to the project plans. Also, **dedicate a slide to the methodologies you should use** to execute the two projects.	25
Pw207	**Quality**! The overall quality of your work will be evaluated. In this activity, all delivered elements will be qualified. **If one or more elements are missing, the score for this element will be zero**.	10
	Total points:	100

5.3 Delivered resources

Besides this document, a series of other resources will be available for you. Here is the overview:

Folder **elements**:

- Subfolder **images**: Images of systems and objects and logotypes, involved in this practical work.
- Subfolder **photoVideo**: Photos and videos of all kind, concerning the railway in general
- Subfolder **templates**:
    - Templates of models for application yEd. Could be used to create the process models for the purchase procedure.
    - Template for KPI specification in Microsoft Excel and Mac Numbers format. Use this file to create your KPIs for element Pw205.
- Subfolder **planning**: A template for the planning to prepare in element **Pw204**. It contains a file in the GanttProject application format which has already pre-defined the roles and resources, as well as the project’s start date. Use this file as a base for your GANTT chart.
- Subfolder **trackPlans**: Contains the track plans, old and new, for the Juberri Ale branch line. I have even added the original file **CargoCare_JuberriAleBranchLine.layout** which can be opened with application “*Railmodeller Pro*”, version 6.6. This is “PayWare”.

All these files, and your own ones, can be used freely in the realization of your delivery.

5.4 Your documents to deliver

Deliver elements Pw201 - **Pw206** in a **zip-archive,** so no jars, rars or other bizarre formats :-). The documents in the zip-file must be both native and exported ones. Create PDF files for export and the native files, created with the tools, like yEd and GanttProject. Structure your files in folders and add a “readme” text file.

I really ask you to create a comprehensive structure in your delivery. Add a **readme** file where you present the contents. This facilitates the location of all your realized work.

Give each file a code with the element’s prefix. Give your files a name like “Pw2_YourNameInCamelCase”. Avoid characters with accents or other bizarre signs. Avoid any risk of inconvenience for transferring the files.

6. Guidelines and hints

In general:
This practical work is very close to a real scenario. If you realize it with success, you will have an interesting experience which serves in your future missions.

Quality:
Create quality! The more you work out the answers, the better you will arrive at the right solution. Each element will be graded, taking into account the quality. **Remember that an incomplete delivery, like one or more missing elements, will set the quality score to zero**.

Don’t use web-tools:
Create **native files** which can be read without an internet connection. Referencing to web-based tools or documents is not allowed and will not be accepted.

BPMN-2 diagram:
Apply a high level of detail. Include Events, Gateways, Sub-processses/Tasks, Exceptions, comments and Data objects. Use the code and title, provided in this document. Please, use yEd for the BPMN-2 diagram so I can analyze it.

The project planning:
Use all the options you have to make clear what is going to happen in the projects. Avoid complexity, specially with dependencies. We want to see the dependency between the activities. Refer to the activity codes to label your tasks in the GANTT diagram.

The project planning tool:
Use GanttProject for the planning, so I can analyze your solution in detail.

Human Resources:
Assign each human resource with the specialization code as a prefix and a sequence number. An example: **`RHCON2`** is the second resource of the construction group. The resources are already defined in the planning template.

To decorate your diagrams and documents:
As for the first practical work, The logotype of “Ferrocarrils Andorrans” is available in the course’s files in topic “**Practical exercise Pw1**”. Use and modify it as you please.

Doubts?
If you have doubts or want to ask a question, please contact me via e-mail: abeening@uda.ad . I can help you with details about the service to put into place. The modeling and presentation is up to you.

7. Conclusion

This exercise is a prototype of a case which could happen in the real world, during your professional career. In this particular case, Ferrocarrils Andorrans will be very happy with a person like you who takes the initiative to prepare and perform these projects!

Enjoy the exercise and thank you really for your time and dedication.

Success and all the best,

Your teacher

Alexander Beening Jansen